

Shifting the Parent Functions

Students will describe and apply vertical and horizontal shifts of the parent functions, and state the domain and range of the shifted graph.

 Graded online lesson • 1.2: [Graphical Transformations](#)

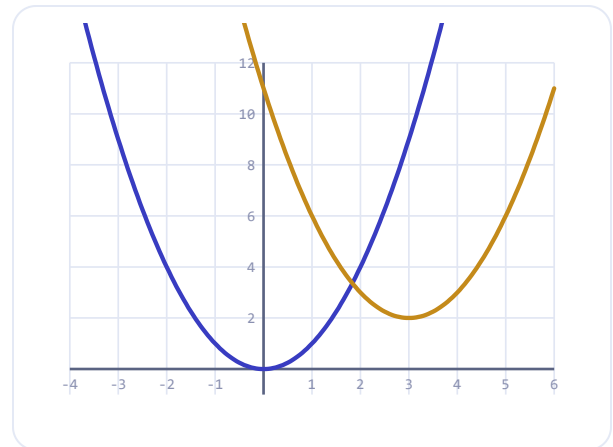
01 • PROJECT Launch deck — Shifting the Parent Functions

HOOK

1 / 7

Same U, new address

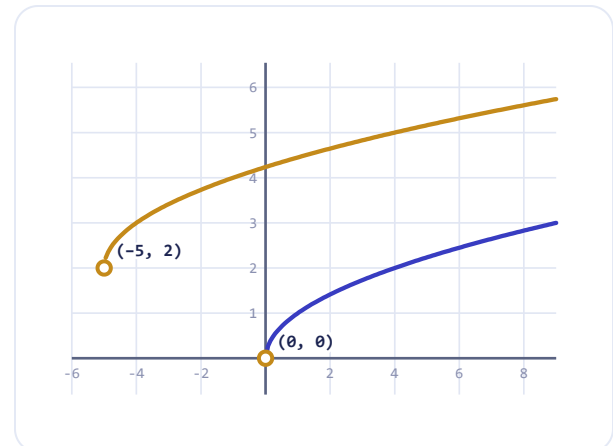
- ◆ Both of these are $y = x^2$ — **identical shape**.
- ◆ One of them just... moved. Where did it go, and what in the equation moved it?
- ◆ By the end of class you'll slide ANY parent anywhere on the grid.



$y = x^2$ (indigo) and its twin (gold) — same shape, new address

The two shift rules

- ◆ **Outside** the function → vertical, **as written**:
 $f(x) + k$ up, $f(x) - k$ down.
- ◆ **Inside** the function → horizontal, **opposite**:
 $f(x - h)$ RIGHT, $f(x + h)$ LEFT.
- ◆ The shape never changes — only the address.



$y = \sqrt{x}$ slid left 5, up 2 — the start point carries the whole graph

Today's words

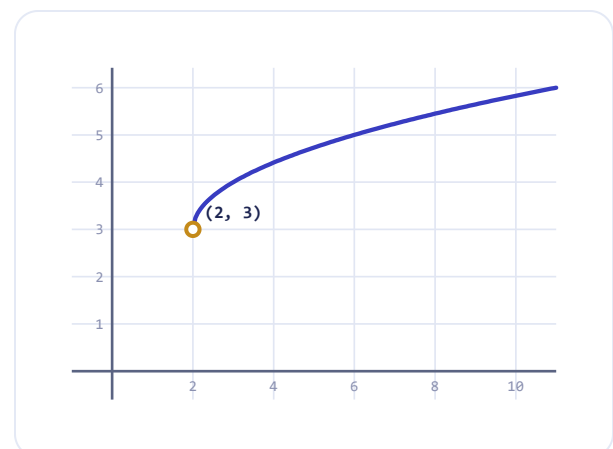
Transformation — any change to a parent's graph

Translation — a slide — shape and size unchanged

We do: $y = \sqrt{(x - 2)} + 3$

WORKED EXAMPLE

1. Inside: $x - 2 \rightarrow$ RIGHT 2 (inside is opposite).
2. Outside: $+3 \rightarrow$ UP 3 (outside is as written).
3. Start point: $(0, 0) \rightarrow (2, 3)$.
4. Domain $x \geq 2$; range $y \geq 3$ — they slide with the graph.



$y = \sqrt{(x-2)} + 3$

The inside sign lies

COMMON MISTAKE

$y = (x - 3)^2$ moves **RIGHT** 3 — not left.

The inside asks "what x makes this zero?" $x = 3$ does, so the vertex now lives at $x = 3$.

KEEP IN MIND

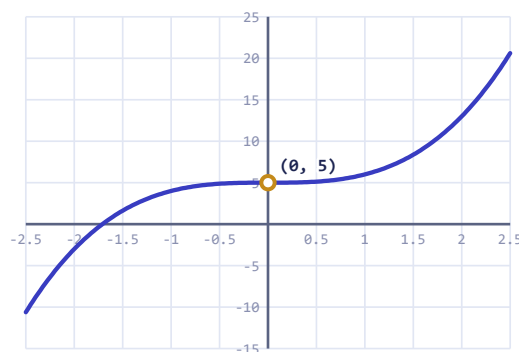
Read the inside as: the value that makes it zero is the new home.

YOU TRY

Your turn

YOU TRY

- Describe every shift: $y = x^3 + 5$
- Write the equation: $y = |x|$ shifted right 3



(a) to check against

WHERE WE'RE HEADED

Where this week goes

- ◆ **Tue** — reflections and stretches (the a out front).
- ◆ **Wed** — combine moves + write equations from a graph.
- ◆ **Thu** — graph any transformed parent from scratch.
- ◆ **Fri** — mixed practice + quiz-style check.

Worked procedure — “I do”

1 Describe every shift of $y = |x + 2| - 5$.

1. Inside: $x + 2 \rightarrow$ LEFT 2 (opposite of the sign).
2. Outside: $-5 \rightarrow$ DOWN 5 (as written).
3. Corner moves $(0, 0) \rightarrow (-2, -5)$.

ANSWER

Left 2, down 5; corner at $(-2, -5)$.

2 Write the equation: $y = \frac{1}{x}$ shifted right 4 and down 3.

1. Right 4 $\rightarrow$ inside becomes $x - 4$.
2. Down 3 $\rightarrow$ append -3 outside.
3. Equation: $y = \frac{1}{x - 4} - 3$.

ANSWER

$$y = \frac{1}{x - 4} - 3$$

3 State the domain and range of $y = \frac{1}{x - 4} - 3$.

1. The vertical asymptote moves to $x = 4$: domain $x \neq 4$.
2. The horizontal asymptote moves to $y = -3$: range $y \neq -3$.

ANSWER

Domain $x \neq 4$; range $y \neq -3$.

Guided notes — teacher key (students get blanks, printed 2-up)

Shifting the Parent Functions

TEACHER KEY • Aug 24

Formula bank **Outside (up/down):** $f(x) + k$ up k · $f(x) - k$ down k — as written

Inside (left/right): $f(x - h)$ RIGHT h · $f(x + h)$ LEFT h — opposite **Domain & range:** slide WITH the graph

VOCABULARY

Transformation — any change made to a **parent function**'s graph

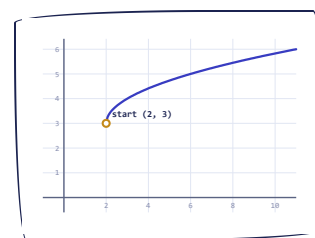
Translation — a **slide** — the shape and size stay exactly the same

EXAMPLE 1 • TYPE 1 — DESCRIBE THE SHIFT

Describe every shift of $y = \sqrt{x - 2} + 3$ and name the new start point.

1. Inside: $x - 2$ moves the graph **right 2** (inside is opposite).
2. Outside: $+3$ moves it **up 3** (outside is as written).
3. The start point $(0, 0)$ lands at **$(2, 3)$** .

Right 2, up 3; start point $(2, 3)$.



EXAMPLE 2 • TYPE 2 — WRITE THE EQUATION

Write the equation: $y = |x|$ shifted left 4 and down 1.

1. Left 4 goes INSIDE with the opposite sign: $|x + 4|$.
2. Down 1 goes OUTSIDE as written: -1 .
3. Equation: $y = |x + 4| - 1$

$$y = |x + 4| - 1$$

EXAMPLE 3 • TYPE 3 — DOMAIN & RANGE AFTER THE SHIFT

State the domain and range of $y = \sqrt{x - 2} + 3$.

1. The graph starts at $(2, 3)$ and runs right and up, so the domain is $x \geq 2$.
2. The lowest output is the start height: range $y \geq 3$.

Domain $x \geq 2$; range $y \geq 3$.

YOU TRY

1.

Describe every shift of $y = x^3 + 5$.

Up 5 only; the center point moves to $(0, 5)$.

2.

Write the equation: $y = \sqrt{x}$ shifted right 6 and down 2, then give its domain and range.

$$y = \sqrt{x - 6} - 2; \text{ domain } x \geq 6, \text{ range } y \geq -2.$$

SUMMARY

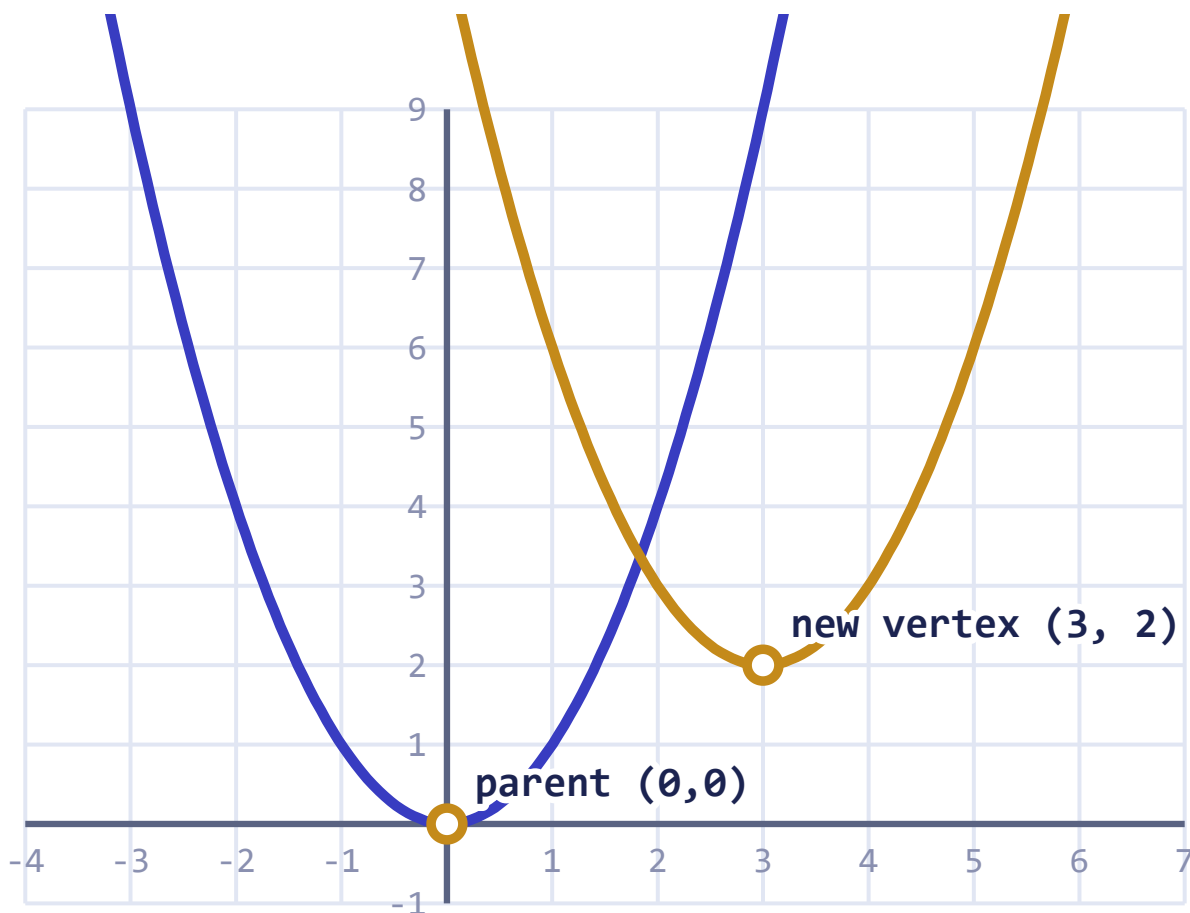
Inside the function moves **opposite**; outside moves **as written**. The domain and range **slide with the graph**.

04 · PRACTICE **Worksheet — Shifting the Parent Functions**

Use the two shift rules: outside moves as written, inside moves opposite. Domains and ranges slide with the graph.

TYPE 1 Describe the shift**EXAMPLE**

$y = (x - 3)^2 + 2$ — inside $x - 3$ moves RIGHT 3, outside $+2$ moves UP 2. The vertex lands at $(3, 2)$.



$y = x^2$ (indigo) vs $y = (x-3)^2 + 2$ (gold)

Directions: Name the parent, then describe every shift.

1. $y = x^2 + 3$

2. $y = \sqrt{x - 4}$

3. $y = |x + 2| - 5$

4. $y = \frac{1}{x} + 6$

TYPE 2 Write the equation

Directions: Write the equation for each described shift.

5. $y = x^3$ shifted down 7

6. $y = |x|$ shifted right 3

7. $y = \frac{1}{x}$ shifted right 4 and down 3

8. $y = \sqrt{x}$ shifted left 5 and up 2

TYPE 3 Domain and range after the shift

Directions: State the domain AND range of each shifted function.

9. $y = \sqrt{x - 4}$

10. $y = \frac{1}{x - 4} - 3$

11. $y = |x + 2| - 5$

CHALLENGE Stretch problems (optional)

12. The point $(1, 1)$ sits on $y = \sqrt{x}$. Exactly where does it land on $y = \sqrt{x + 5} + 2$? Explain both moves.

05 • KEY Answer key**DESCRIBE THE SHIFT**

1. Quadratic parent; up 3 (vertex $(0, 3)$)
2. Square-root parent; right 4 (start $(4, 0)$)
3. Absolute-value parent; left 2 and down 5 (corner $(-2, -5)$)
4. Reciprocal parent; up 6 (horizontal asymptote $y = 6$)

WRITE THE EQUATION

5. $y = x^3 - 7$

6. $y = |x - 3|$

7. $y = \frac{1}{x - 4} - 3$

8. $y = \sqrt{x + 5} + 2$

DOMAIN AND RANGE AFTER THE SHIFT

9. Domain $x \geq 4$; range $y \geq 0$

10. Domain $x \neq 4$; range $y \neq -3$

11. Domain: all real numbers; range $y \geq -5$

CHALLENGE

12. $(1, 1) \rightarrow (-4, 3)$: inside $x + 5$ slides left 5 ($1 \rightarrow -4$), outside $+2$ slides up 2 ($1 \rightarrow 3$)

06 · CLOSE Exit ticket

1. Describe every shift of $y = \frac{1}{x + 5} - 2$, then give its domain and range.